WEB APPLICATION DEVELOPMENT FOR DETECTING POTENTIALLY HARMFUL DRUG INTERACTIONS IN MEDICAL PRESCRIPTIONS

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This thesis discusses the development of a web application to detect potentially dangerous drug-drug interactions in patient prescriptions. The research is motivated by the current practice at ITB's Primary Clinic, where the drug interaction verification process is still done manually—relying on senior pharmacists' memory or junior pharmacists' use of textual reference applications. This method is considered inefficient and prone to errors, especially when prescriptions involve many drugs.

This study aims to: (1) design and implement a web-based drug interaction verification system, and (2) evaluate the system's performance in terms of time efficiency and memory usage by applying various data searching methods. Development followed the waterfall model, utilizing PHP, MySQL, HTML, CSS, and JavaScript. The drug-drug interaction dataset was compiled from valid medical references and used to detect interactions based on the drug names entered.

Testing was conducted using two approaches: system functional testing and user trials. System testing ensured that features worked as designed. User trials involved doctors, pharmacists, and administrators of the ITB Primary Clinic, as well as other medical personnel, to assess the system's ease of use and its alignment with the clinic's workflow. Additionally, the performance of various data searching methods (CSV, SQL, dictionary) was tested for execution time and memory usage in scenarios involving 5, 20, 50, and 100 drugs.

he test results show that the system can detect drug interactions accurately and efficiently. The system was rated as easy to use, helpful for medical personnel's workflow, and has the potential to be integrated into electronic medical record systems. The database lookup table-based data structure approach proved to be efficient for large-scale data processing.

Keywords: drug interactions, patient prescriptions, web application, verification system, service efficiency, clinical pharmacy, medication safety.

I. INTRODUCTION

The advancement of information technology in the healthcare sector has driven numerous innovations aimed at improving the efficiency and quality of medical services. One critical area that requires special attention is pharmaceutical practice, particularly in ensuring the safety of concurrent drug use. Drug-drug interactions (DDIs) pose a significant challenge, especially in patients undergoing polypharmacy—those who consume multiple medications to manage various conditions simultaneously. This situation is common among individuals with chronic illnesses such as cancer, diabetes, and cardiovascular diseases, as well as in elderly populations experiencing physiological changes due to aging (Xiong et al., 2022; Sitar, 2007).

Drug interactions can lead to undesirable and harmful pharmacological effects, including reduced therapeutic efficacy, toxicity, or even adverse drug reactions (ADRs). While some interactions may be synergistic and beneficial, most clinical DDIs are negative and pose a serious threat to patient safety (Xiong, 2021). Studies indicate that the risk of DDIs increases proportionally with the number of medications taken. Approximately 40% of patients using five drugs face a high risk of serious interactions, a figure that rises to over 80% for those taking seven or more medications (Grattagliano et al., 2010).

Multiple reports have shown that DDIs contribute to a rise in ADR incidences, emergency department visits, hospitalizations, and rehospitalizations. A study in Romania revealed that 25.9% of all validated ADRs were caused by drug interactions (Farcas et al., 2010). In Indonesia, this issue has yet to be addressed optimally. Documentation and monitoring of DDIs in healthcare facilities remain limited, and clinical decision support systems (CDSS) specifically designed to verify drug interactions are still rarely developed. Local studies indicate that 63.5% of analyzed prescriptions have potential drug interactions, most of which are moderate, with some classified as major and potentially resulting in serious clinical consequences (Reyaan et al., 2021; Indriani & Oktaviani, 2020).

Interviews with pharmacists from various healthcare facilities, such as Klinik Pratama ITB and the University of Indonesia Hospital, reveal that drug interaction checks are generally performed manually. Senior pharmacists often rely on experience and memory, while existing support tools like Micromedex are considered impractical for daily use due to the need to search for each drug name individually. Additionally, official data from institutions such as the Indonesian Food and Drug Authority (BPOM) is still provided in static formats (PDF), which do not support fast and interactive searches.

This situation highlights the urgent need for an information system that can support the drug interaction verification process in an efficient, accurate, and user-friendly manner. Such a system is expected to provide a practical solution to assist healthcare professionals in identifying potential drug interactions, while simultaneously enhancing patient safety through more structured and responsive data-driven decision-making. By leveraging adaptive technology that integrates with clinical workflows, this system can serve as a crucial foundation for promoting safer, evidence-based pharmaceutical practice ease of use.

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A. *Problem Formulation*

Based on the background and the focus that has been described, the following are the problems that will be discussed by the author in this final project:

1. How to design and implement a web-based drug interaction verification system that can automatically detect drug-to-drug interactions based on the drug names listed in patient prescriptions?
2. How is the system's performance in terms of processing speed and memory usage in the drug interaction verification process based on the test results?Maintaining the Integrity of the Specifications

B. *Objective*

The objectives of this final project are as follows:

1. To design and implement a web-based drug interaction verification system that can automatically detect potential drug-to-drug interactions based on the drug names listed in patient prescriptions.
2. To evaluate the performance of the developed system in terms of processing speed and memory usage during the drug interaction verification process.

II. Prepare Your Paper Before Styling

A. *Types of Drugs and Their Use in Indonesia*

Drugs are defined as chemical or biological substances used to influence physiological systems or pathological conditions, whether for diagnostic, preventive, therapeutic, or health improvement purposes (BPOM, 2024). In Indonesia, drugs are classified into two main categories: generic drugs, which use standardized active ingredient names, and branded/patented drugs, which are marketed under specific trade names and protected by patents for up to 20 years (Puspita & Rissa, 2023). Although generic drugs are more cost-effective, their usage rate in Indonesia remains low (approximately 7%) due to public perceptions that they are of lower quality compared to branded drugs (Alim, 2018).

B. *Adverse Drug Reactions (ADR)*

According to the WHO, an Adverse Drug Reaction (ADR) is an unintended and harmful response to a drug administered at normal doses for prophylaxis, diagnosis, or therapy. ADRs are classified into two main mechanisms: immunological and non-immunological reactions. The majority of ADRs (up to 80%) fall under the non-immunological category, which includes predictable pharmacological effects (e.g., hypoglycemia due to insulin), idiosyncratic reactions, and drug-drug interactions (DDIs) (Riedl & Casillas, 2003).

C. *Drug-Drug Interactions (DDI)*

Drug interactions occur when one drug alters the pharmacological effect of another, either through pharmacokinetic mechanisms (absorption, distribution, metabolism, excretion) or pharmacodynamic mechanisms (altering effects without changing pharmacokinetics) (Xiong et al., 2022). DDIs are a common cause of predictable (Type A) ADRs and are a major concern in polypharmacy cases, as they may reduce therapeutic effectiveness and compromise patient safety (Bucşa et al., 2013).

D. *System Architecture*

A three-tier architecture is commonly used in the development of web-based health applications and consists of the following layers:

1. Presentation Layer – Responsible for the user interface, typically built with HTML, CSS, and JavaScript.
2. Business Logic Layer – Executes application logic, usually implemented in languages such as Python, PHP, or Java.
3. Data Access Layer – Manages data storage and retrieval from databases through CRUD (Create, Read, Update, Delete) operations.

E. Database Design and Indexing

The system uses a Relational Database Management System (RDBMS) such as MySQL to manage structured medical data. MySQL supports referential integrity, SQL queries, and indexing with B+ Tree structures to optimize search performance. The use of indexes is critical to avoid full table scans and to accelerate data access, especially in large datasets (Silberschatz et al., 2010; Elmasri & Navathe, 2015).

F. *Data Search Techniques*

Various data retrieval techniques are employed in the drug interaction verification system, considering factors such as speed, memory efficiency, and scalability:

1. Simple Dictionary (Hash Map) – Offers extremely fast (O(1)) single-level key-value lookups.
2. Nested Dictionary – Supports multi-dimensional data searches but is more complex and memory-intensive.
3. Pandas MultiIndex – Enables multi-key indexing in tabular DataFrames, suitable for large-scale data but requires significant memory and external libraries.
4. Indexed SQL Search – Utilizes database indexing (e.g., B+ Tree) to accelerate large-scale searches.
5. Manual CSV Parsing – Reads CSV files line-by-line; simple but inefficient for large datasets due to linear access time (O(n)).

Each technique presents its own advantages and limitations in terms of access speed, memory efficiency, and implementation complexity—critical factors in developing a real-time drug interaction verification system.

design and implement a web-based drug interaction verification system that can automatically detect potential drug-to-drug interactions based on the drug names listed in patient prescriptions.

III. METODOLOGY

Overall, the methodology applied in the design of this final project can be divided into seven processes: interviews and discussions, problem formulation, analysis of specifications and solution requirements, system specification development, implementation of system design and specifications, system testing, and finally, evaluation of the system results. The following is the flowchart of the system development methodology.

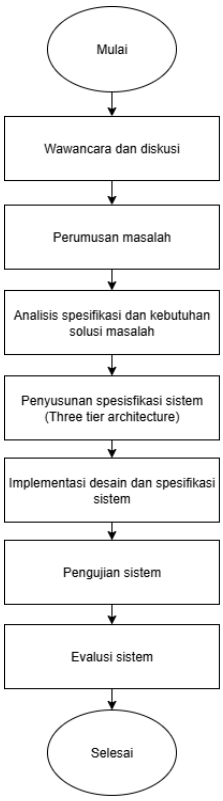

Fig. 1. Metodology

A. User Requirements Description

The system application design is intended to meet the needs of three main categories of users: doctors, pharmacists, and administrators. The following table shows the tasks and access rights of each user:

Based on the characteristics and interviews with doctors, pharmacists, and administrators at the primary clinic, the following user requirements were identified:

TABLE I. DOCTORS REQUIREMENT

Code	Requirement
US-01	Doctors need to log in to the application.
US-02	Doctors can prescribe medication by entering the drug name (generic/brand).
US-03	Doctors can view the results of drug interaction checks for prescribed drugs.
US-04	Doctors can edit prescriptions if dangerous interactions are found.
US-05	Doctors can access prescription history on the same day.

TABLE II. PHARMACY REQUIREMENTS

Code	Requirement
US-06	Pharmacists need to log in to the application.
US-07	Pharmacists can view the doctor's prescription and the drug interaction results.
US-08	Pharmacists can add other drugs (outside the prescription) to check for interactions.
US-09	Pharmacists can access the patient's prescription history for the past 5 years in accordance with regulations.

TABLE III. ADMIN REQUIREMENTS

Code	Requirement
US-10	Admins need to log in to the application.
US-11	Admins can add user accounts (doctors/pharmacists).
US-12	Admins can delete user accounts.
US-13	Admins can update the drug database (generic/brand).

B. Non Functional Reequirement

The system application design is structured to meet the needs of three main user categories: doctors, pharmacists, and administrators. The table below outlines the tasks and access rights of each user group.

Based on characteristics and interviews with doctors, pharmacists, and administrators at the primary clinic, the following user requirements were identified:

These requirements are directly related to the processes that the system will perform. All the formulated user requirements are functional requirements, which represent the main features and services that the system must provide to ensure optimal usability for users (doctors, pharmacists, and administrators). The list of functional requirements derived from the user needs analysis is presented in the following table:

TABLE IV. FUNCTIONAL REQUIREMENTS

ID	**Requirement Description**
F-01	The system provides a registration feature
F-02	The system provides a login feature
F-03	The system provides a logout feature
F-04	The system restricts access based on user roles
F-05	The system provides a profile viewing feature
F-06	The system provides a profile editing feature
F-07	The system provides a password change feature
F-08	The system provides a feature to input patient data and diagnosis
F-09	The system provides a prescription input feature
F-10	The system provides a drug interaction checking feature
F-11	The system provides a feature to edit prescription
F-12	The system provides a feature to view today's prescription history and detailed view
F-13	The system provides a feature to add drugs to check additional interactions

F-14	The system provides a feature to view a patient's prescription history over 5 years
F-15	The system provides a feature to add drug data
F-16	The system provides features to view, edit, and delete application user accounts

Fig. 2. Use Case Diagram

TABLE V. KEBUTUHAN NONFUNGSIONAL

ID	**Parameter**
NF-01	*Performance*
NF-02	*Reliability*
NF-03	*Readability*
NF-04	*Interface*
NF-05	Batasan Data

Performance

The application is designed to run across various devices through a web-based platform. Users are expected to be able to switch pages, load data, and submit information quickly and efficiently without interruptions or delays. This aims to create a smooth user experience, especially during critical processes such as prescription writing, data verification, and patient information management.

Reliability

System reliability emphasizes the security of user data. Sensitive information such as patient identities, medical histories, and prescription data must be well protected. The application should be available at all times without technical disruptions. The system enforces access restrictions based on user roles—such as doctors, pharmacists, and administrators—so that each user can only access features relevant to their responsibilities. This is crucial to prevent data breaches and maintain system integrity.

Readability

To ensure ease of understanding for all users, the system uses Bahasa Indonesia as the primary language in its interface and documentation. This reduces the risk of misunderstanding in system usage and facilitates adoption by medical personnel who may not be familiar with technical terms in foreign languages.

Interface

The user interface design focuses on ease of use and intuitiveness. Users do not need to undergo complex training to understand and operate the application. A simple and consistent system navigation helps speed up workflows, reduce errors, and improve operational efficiency in clinical environments.

Data Constraints

The system must be capable of storing and managing large volumes of data over time without compromising application performance. High storage capacity and optimized data management are required to ensure the application remains responsive as the data volume continues to grow.

C. *Database Design*

TABLE VI. DATABASE DESIGN

able Name	Description
users	Stores user account data and is related to the roles table.
roles	Stores user roles such as admin, doctor, or pharmacist.
patients	Stores patient information (name, date of birth, address, and diagnosis).
doctors, pharmacists, admins	Store specific data based on the user's role, linked to user_id.
prescriptions	Stores prescription information from doctors, related to doctors and patients.
prescription_items	Stores detailed medication information in a prescription, related to master_nama_obat.
master_nama_obat	Stores the list of drug names used in the system.
drug_ingredients	Represents the relationship between drugs and their active ingredients, linking master_nama_obat to master_zat_aktif.
master_zat_aktif	Stores a list of active pharmaceutical ingredients (APIs).
data_interaksi_indexed	Stores indexed interaction data between active ingredients, with id_1 and id_2 referencing master_zat_aktif.

Fig. 3. *Entity Relation ship Diagram*

The figure above illustrates the relationships between the database tables used in the system.

These relationships represent how entities such as users, doctors, pharmacists, patients, and administrators are connected to prescription data, medications, and their active components. The users table acts as the primary entity for managing user authentication and authorization, with a relationship to the roles table to differentiate the roles of each user in the system.

Each user with the role of doctor, pharmacist, or administrator has a one-to-one relationship with the doctors, pharmacists, and admin tables respectively. The prescriptions table stores prescription data, which is related to doctors and patients, while the specific medications within a prescription are recorded in the prescription_items table. This table is then linked to the drug information stored in master_nama_obat.

The master_nama_obat table stores the names of medications, which are then linked to the drug_ingredients table to represent the active pharmaceutical ingredients (APIs) contained in each drug. These active ingredients are represented in the master_zat_aktif table, which is subsequently used in the interaction-matching process through the data_interaksi_indexed table.

D. *Search Method Design*

In this study, six methods for searching drug-drug interaction data were implemented and compared. These methods are grouped into two main categories based on the data processing approach used: File-Based Processing and Database Query Processing.

1. File-Based Processing

This approach utilizes files as the primary data source, with all interaction data loaded into memory (RAM) during the search process. The four main methods in this category are:

Simple Dictionary

Utilizes a flat dictionary data structure to store active ingredient pairs and their interaction levels as key-value pairs. This method offers a simple yet efficient approach for relatively small datasets.

1. Nested Dictionary

Employs a hierarchical dictionary structure where each active ingredient serves as a primary key containing another dictionary as a secondary key. The inner dictionary maps to the interaction level. This supports efficient bidirectional lookup.

2. MultiIndex Pandas (Memory-Based)

Leverages the MultiIndex feature of the Pandas library to store interaction pairs in a DataFrame with multi-level indexing. All interaction data is loaded and held in memory throughout the search process. This method is suitable for large datasets that still fit within system memory.

3. MultiIndex Pandas (File-Based)

A variant of the previous method, but with the key difference that the indexed data is not retained in memory. After indexing, the data is saved to a file (e.g., Pickle or JSON format), and the main process performs lookups directly from the file. This method is designed to reduce memory usage and enhance efficiency in resource-constrained environments.

4. Database Query Processing

In this approach, all interaction data is preprocessed and stored within a relational database system (MySQL). Tables such as master_nama_obat, drug_ingredients, and data_interaksi_indexed contain preprocessed data rather than raw input. These tables are indexed to optimize query performance.

Unlike file-based methods, this approach does not load the entire dataset into memory. Instead, it relies on SQL commands to selectively retrieve the required data. This makes it well-suited for large-scale production systems that utilize database management systems as the backbone for data storage and retrieval.

IV. RESULT

Functional Testing Results

1. The testing flow using the Black Box Testing approach was carried out through the following five stages:
2. Identifying the system's functional requirements based on the Functional Requirements Table (Table III.4);
3. Developing tasks or test scenarios that represent each requirement;
4. Identifying test procedures for each scenario;
5. Determining the expected output of each scenario;

TABLE VII. UJI FUNCTIONAL RESULT

No	Test Scenario	Test Procedure	Code	
1	Account registration	1. Access the registration page2. Enter user data3. Click the “Register” button	F-02	✓
2	Perform login	1. Access the login page2. Fill in username and password3. Click the “Login” button	F-03	✓
3	Perform logout	Click the logout button on the navbar/menu	F-04	✓

4	Access restriction by role	1. Login as each role2. Attempt to access pages not authorized for that role	F-05	✓
5	View user profile	1. Click on the "Profile" menu2. View user information	F-06	✓
6	Update user data (Edit Profile)	1. Dashboard → Profile2. Click “Edit Profile”3. Fill in the form4. Click “Save”	F-07	✓
7	Change password	1. Profile → Click “Change Password”2. Fill in old and new passwords3. Click “Change Password”4. Test incorrect old password5. Test mismatched new password confirmation	F-08	✓
8	Input prescription	1. Open prescription input page2. Fill in patient and prescription data3. Click “Submit and Check Interaction”4. Ensure submission works with only one prescription item	F-09	✓
9	Perform interaction check	1. System automatically checks interaction after input2. Display result if interactions found3. Display result if no interactions found	F-10	✓
10	Edit prescription if interaction is found	1. Open prescription history2. Click “Edit” on the problematic prescription3. Save changes	F-11	✓
11	View today’s prescription history	Pharmacist views today’s prescription history and its details	F-12	✓
12	Add additional medication (check for additional interactions)	1. Pharmacist clicks “Add Medication” on the prescription detail page to check non-prescribed meds2. System checks additional interaction3. Ensure the original prescription remains unchanged	F-13	✓
13	Add drug data (update database)	1. Open add/update drug data page2. Upload CSV file	F-14	✓

The testing activities were conducted directly on Friday, July 25, 2025, at the Klinik Pratama ITB. Each system role was tested by one representative from Klinik Pratama ITB according to their respective functions. In addition, to broaden the perspective and validate the results, the testing also involved several additional participants: two clinical rotation (co-assistant) doctors, one pharmacy student from the University of Indonesia with internship experience at RSUI (Universitas Indonesia Hospital), and two pharmacy students from Institut Teknologi Bandung.

The limitation in the number of respondents was due to constraints in time and available resources. However, the selection of participants was carried out through **purposive sampling**, based on specific criteria such as professional background, experience, and the relevance of their roles to the system. Therefore, the testing results are still considered **representative** in illustrating real-world usage of the application in a clinical setting.

All participants were first given a **general briefing** about the system, including an explanation of the interface structure, usage flow, and the main functions of each feature. After gaining a basic understanding, the participants were asked to follow a series of **testing scenarios** that had been designed according to their respective roles.

In the final stage of the testing, all participants were asked to complete a **user evaluation questionnaire** via an online form. The evaluation instruments used in this study consisted of **three main components**:

1. **System Usability Scale (SUS)** – Used to measure users' perceptions of the overall ease of use of the system. SUS is a widely adopted and validated instrument for quantitatively assessing system usability.
2. **Single Ease Question (SEQ)** – Used to assess the user's perceived ease in completing specific tasks, such as entering prescriptions or checking for drug interactions. SEQ provides a direct insight into user experience when performing particular functions.
3. **In-Depth Interview (IDI)** – Conducted qualitatively using a semi-structured approach to explore in more depth the users' opinions, experiences, encountered obstacles, and suggestions or feedback on the tested system.

TABLE VIII. SUS VALUE

Number of The Question Code	Value
QS-1	5
QS-2	1,14
QS-3	4,86
QS-4	1,14
QS-5	5
QS-6	1
QS-7	4,57
QS-8	1,14
QS-9	4,86
QS-10	1
Hasil Perhitungan SUS	97,14

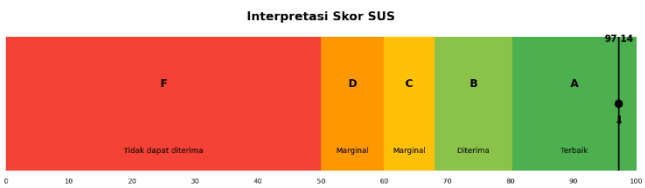

Fig. 4. SUS Result

Based on the calculation results, the system achieved a SUS score of 97.14, placing it in the "A" or "Excellent" category according to the standard SUS interpretation, as illustrated in the figure below the table. According to Brooke (1996), a score above 85 indicates a system with very high usability and can be interpreted as a "highly usable" product. This interpretation is further supported by the interpretive framework developed by Bangor et al. (2009), which states that a SUS score in the range of 90–100 reflects exceptional user satisfaction, with strong potential for widespread use without the need for additional training. Therefore, it can be concluded that the developed system has fully met the usability aspect and provides an optimal user experience.

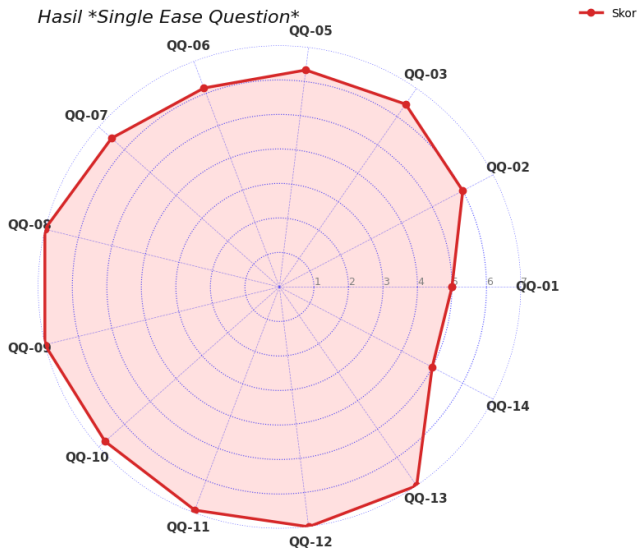

Fig. 5. SEQ Result

The radar chart representing the Single Ease Question (SEQ) results shows an almost evenly distributed pattern, approaching the maximum score values. This indicates a consistent level of ease across various tested features. Although there were one or two tasks with slightly lower scores — such as account registration and account deletion (both scoring 5) — this does not indicate significant usability issues, but rather reflects the limited usage context, especially for features accessed only by the administrator role. In this case, the SEQ scores of 5 were given by the sole respondent in the administrator role, and therefore reflect individual evaluation based on actual experience, not aggregate results.

The in-depth interviews (IDI) conducted with users from the three categories — doctors, pharmacists, and administrators — revealed highly positive responses toward the drug interaction verification application. Overall, users expressed that the system is easy to use, supports their daily tasks, and has a clean, professional interface.

Key strengths highlighted include:

- The real-time drug interaction check feature during prescription writing;
- The flexibility to input both generic and brand-name drugs, which adds great value;
- A calm, informative interface and intuitive navigation appreciated by all respondents.

From a functionality perspective, features such as daily prescription lists, detailed patient prescriptions, and active ingredient interaction checks were considered highly helpful, especially by pharmacists, in enhancing patient safety.

Constructive feedback provided includes:

- Adding important fields such as body weight, allergy information, and formatting prescriptions according to medical standards;

- Including digital signatures for legal validation by doctors;
- Expanding the list of dosage forms and doses to improve prescription efficiency;
- Providing interaction severity levels (major, moderate, minor) for educational purposes;
- Introducing a drug catalog and brand-name/substitution references for ease of use.

No major usability issues were reported. All features were deemed relevant, and none of the respondents expressed confusion in navigation after the initial briefing.

In conclusion, the IDI results confirm that the system meets fundamental functional requirements, is well-received by users, and holds strong potential for future development.

V. CONCLUSION AND RECOMMENDATION

A. Conclusion

1. System Successfully Implemented

The web-based drug interaction verification system has been successfully designed and implemented using a three-tier architecture (client, application server, and database). This system can automatically detect drug interactions based on both generic and brand drug names in patient prescriptions, supported by data from BPOM and DDInter.

2. Optimal Performance with PHP Database Lookup

Performance evaluation shows that the PHP Database Lookup method is the most efficient and balanced approach in terms of speed and memory consumption. This method is suitable for medium to large-scale web systems.

B. Recommendations and Suggestions

1. Integration of Physical Prescription Upload and OCR

To accommodate patients with prescriptions outside the system, it is recommended to add a feature for uploading prescription images equipped with Optical Character Recognition (OCR) capabilities.

2. Standardization of Prescription Format & EMR Integration

Prescription forms should comply with Electronic Medical Record (EMR) standards, including medical record numbers, allergy information, and patient anthropometric data.

3. Interaction Check Based on Medication History

The system should be able to check drug interactions based on the patient's medication history.

4. Automatic Drug Stock Availability Check

Add a feature to check drug availability so that only available drugs are shown during prescription input.

5. Recording of Medical Consumables and Pricing

A feature to manage medical consumable items (BMHP) and unit pricing should be added for more comprehensive medical management.

6. Drug Catalog and Composition

A drug catalog page should be provided with information on composition, description, and alternative brand names as an additional reference.

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